

Running into error 'Composition is not defined over Operators of different

<https://github.com/qiskit-community/qiskit-aqua/issues/998>

ROW 48

Running into error 'Composition is not defined over Operators of different dimensions, 4 and 8' #998



Closed

#1048



plussoyeur opened on May 18, 2020 · edited by plussoyeur

Edits ...

Information

- Qiskit Aqua version: 0.7.1
- Python version: 3.7.3
- Operating system: NAME= elementary OS / 5.1.4 Hera

What is the current behavior?

I am trying to run a QAOA algorithm and I run into a dimensionality issue when I try to append an initial circuit. If I remove this circuit, everything works perfectly fine. I did check that my initial circuit has 4 qubits and the same holds for the operator Hc.

Steps to reproduce the problem

```
import itertools
import numpy as np
import itertools
from functools import reduce, partial

from qiskit import Aer, QuantumRegister, ClassicalRegister, execute, QuantumCircuit
from qiskit.tools.visualization import circuit_drawer
from qiskit.quantum_info import Pauli
from qiskit.aqua import QuantumInstance
from qiskit.aqua.operators import WeightedPauliOperator
from qiskit.aqua.components.initial_states import Custom
from qiskit.aqua.components.optimizers import COBYLA
from qiskit.aqua.algorithms import QAOA

np.set_printoptions(precision=3, suppress=True)

n_qubits = 2
n_system = n_qubits * 2
T = 1000

weights = np.array([[0,1],[0,0]])
p = 2

def product_pauli_z(q1, q2, coeff):
    eye = np.eye(n_system)
    return WeightedPauliOperator([coeff, Pauli(eye[q1], np.zeros(n_system)) * Pauli(eye[q2], np.zeros(n_system))])

def ising_hamiltonian(weights):
    H = reduce(lambda x,y:x+y,
               [product_pauli_z(i,j, -weights[i,j])
                for (i,j) in itertools.product(range(n_qubits), range(n_qubits))])
    #H.to_matrix()
    return H

Hc = ising_hamiltonian(weights)
print(Hc.print_details())

qr = QuantumRegister(n_system)
cr = ClassicalRegister(n_system)
backend = Aer.get_backend('qasm_simulator')

def prepare_init_state(T):
    init_circuit = QuantumCircuit(qr)
    a = 2 * np.arctan(np.exp(- 1/T))
    for i in range(n_qubits):
        init_circuit.rx(a, qr[n_qubits+i])
        init_circuit.cx(qr[n_qubits+i], qr[i])
    init_state = Custom(n_system, circuit=init_circuit)
    return init_state

initial_state = prepare_init_state(T)
print(prepare_init_state(T).construct_circuit())

optimizer = COBYLA()
qaoa = QAOA(Hc, optimizer, 2, initial_state)
```

Assignees

No one assigned

Labels

type: bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

CircuitOp uses QuantumCircuit.compose, not .combine
qiskit-community/qiskit-aqua

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

Participants



```

initial_state = prepare_init_state(T)
print(prepare_init_state(T).construct_circuit())

optimizer = COBYLA()
qaoa = QAOA(Hc, optimizer, 2, initial_state)

backend = Aer.get_backend('statevector_simulator')
quantum_instance = QuantumInstance(backend)
result = qaoa.run(quantum_instance)

```



Cryoris on May 18, 2020 · edited by Cryoris

Edits Contributor ...

This seems to be a bug caused by naming explicitly creating the qubit register of the initial state, since the composing circuits with differently named registers can stack them on top of each other instead of appending them. Try running this slightly modified version where the qubit register is not explicitly defined

```

def prepare_init_state(T):
    init_circuit = QuantumCircuit(n_system)
    a = 2 * np.arctan(np.exp(- 1/T))
    for i in range(n_qubits):
        init_circuit.rx(a, n_qubits+i)
        init_circuit.cx(n_qubits+i, i)
    init_state = Custom(n_system, circuit=init_circuit)
    return init_state

```

Until we figure out where this composition error comes from.



plussoyeur on May 18, 2020

Author ...

Thanks a lot. This works. I did try that on an older version of qiskit and that did not solve the issue but this works perfectly here.



1

- Cryoris added **type: bug** on May 18, 2020
- mavzolej mentioned this on Jun 11, 2020
 - Another "Running into error 'Composition is not defined over Operators of different dimensions'" #1034
- woodsp-ibm added this to To do in [Aqua Core](#) on Jun 11, 2020
- Cryoris mentioned this on Jun 16, 2020
 - CircuitOp uses QuantumCircuit.compose, not .combine #1048
- Cryoris closed this as **completed** in #1048 on Jun 16, 2020
- Cryoris moved this from To do to Done in [Aqua Core](#) on Jun 16, 2020

CircuitOp uses QuantumCircuit.compose, not .combine #1048

Merged Cryoris merged 4 commits into `qiskit-community:master` from `Cryoris:fix-operator-circuit-compose` on Jun 16, 2020

Conversation 0 Commits 4 Checks 0 Files changed 3

+25 -3



Cryoris commented on Jun 16, 2020

Contributor

Summary

The `compose` method of the `CircuitOp` used `QuantumCircuit.combine` which has been changed to use `QuantumCircuit.compose`.

Fixes [#998](#) and [#1034](#).

Details and comments

Using `combine` leads to the problem that composing an operator with a `CircuitOp` based on a named register does not chain the operators but stacks them. E.g. composing `Z ^ 2` with a circuit based on a 2-qubit named register yielded a 4-qubit operator instead of a 2-qubit operator.

Merge branch 'master' into fix-operator-circuit-compose

Cryoris added 3 commits 5 years ago

use `compose` instead of `combine`

Unverified b08fa8d

Reviewers

woodsp-ibm ✓

manoelmarques ●

Assignees

No one assigned

Labels

Changelog: Bugfix

stable backport potential

Projects

None yet

Milestone

No milestone